05/11/26

Signal Processing

- Recap: Ideal lowpass linear phase system
- Generalized linear phase systems
- Causal GLP systems
  - Types I to IV

- Recap: $h_{lp}[n] = \dfrac{\sin \omega_c (n-\alpha)}{\pi (n-\alpha)}$

If $\alpha = n_d$, an integer delay.

$$h_{lp}[2\alpha - n] = \frac{\sin \omega_c (2n_d - n - n_d)}{\pi (2n_d - n - n_d)} = h_{lp}[n].$$

- Symmetry plays an important role in linear-phase filter design.
- $H(e^{j\omega}) = |H(e^{j\omega})| \exp \angle H(e^{j\omega})$
- Generalized linear-phase (GLP) systems:

$$H(e^{j\omega}) = \underbrace{A(e^{j\omega})}_{\text{real}} e^{-j(\omega\alpha - \beta)} \quad -(1)$$

(α, β: real)

Contrast:

$$H_{lp}(e^{j\omega}) = \begin{cases} 1 \cdot e^{-j\omega\alpha}, & |\omega| < \omega_c \\ 0, & \omega_c < |\omega| \le \pi \end{cases}$$

$$H(e^{j\omega}) = A(e^{j\omega}) \cdot [\cos(\beta - \omega\alpha) + j \sin(\beta - \omega\alpha)] \quad -(2)$$

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} h[n] \cdot (\cos\omega n - j \sin\omega n) \quad -(3)$$

Now what is $\dfrac{\sin(\beta - \omega\alpha)}{\cos(\beta - \omega\alpha)} = \dfrac{-\sum_{n=-\infty}^{\infty} h[n] \cdot \sin\omega n}{\sum_{n=-\infty}^{\infty} h[n] \cos\omega n}$

$$\Rightarrow \left(\sum_{n=-\infty}^{\infty} h[n]\cdot \cos \omega n\right)\cdot \sin(\beta-\omega\alpha) + \left(\sum_{n=-\infty}^{\infty} h[n]\sin\omega n\right)\cdot \cos(\beta-\omega\alpha)$$

$$= 0$$

$$\sum_{n=-\infty}^{\infty} h[n]\left(\cos \omega n\cdot \sin(\beta-\omega\alpha) + \sin\omega n\cdot \cos(\beta-\omega\alpha)\right] = 0$$

$$\sum_{n=-\infty}^{\infty} h[n]\sin\left(\omega(n-\alpha)+\beta\right) = 0$$ (real) (necessary condition for GLP)

- $\beta = 0$ or π
- $2\alpha = M$, an integer
- $h[2\alpha-n] = h[n]$

} Satisfies our GLP condition.

- $\beta = \pi/2$ or $3\pi/2$
- $2\alpha = M$, an integer
- $h[2\alpha-n] = -h[n]$

- Let's bring in the causality condition. $\Rightarrow h[n] = 0$ for $n<0$
- We need $h[M-n] = h[n] \Rightarrow$ we have an FIR system.
- Since $h[n] = 0$ $n<0$, $h[M-n] = 0$ for $n>M$. ($\because h[M-n] = h[n]$)

$$\therefore h[n] = \begin{cases} h[M-n], & 0 \le n \le M \\ 0, & \text{otherwise} \end{cases}$$ $\Rightarrow h[n]$ is FIR with length $(M+1)$.

- Let's say M is even, i.e., we have an odd length filter, we would like a system whose impulse response is $A(e^{j\omega})\cdot e^{-j\omega\alpha + j\beta}$ (real)
- Show that if $h[M-n] = h[n]$ we can indeed write $H(e^{j\omega}) = A(e^{j\omega})\cdot e^{-j\omega\alpha+j\beta}$.

$$H(e^{j\omega}) = \sum_{n=0}^{M} h[n]\, e^{j\omega n} \;;$$

$$= \underbrace{e^{-j\omega M/2}}_{\text{linear phase}} \underbrace{\left(\sum_{n=0}^{M/2-1} h[n]\cdot 2\cdot \cos\omega\left(\frac{M}{2}-n\right) + h[M/2]\right)}_{A_e(e^{j\omega}),\ \text{GLP even, real}}$$

- Example: $h[n] = \begin{cases} \frac{1}{5} & 0 \le n \le 4 \\ 0 & \text{else} \end{cases}$ (Type I)

$= \frac{1}{5}[\delta[n] + \delta[n-1] + \dots + \delta[n-4]]$

averaging filter

- Types of FIR systems:

M	Symmetry	Antisymmetry
Even	I	III
odd	II	IV

- Example: $h[n] = \underbrace{\delta[n] - \delta[n-2]}$ (Type III)

GLP

$$H(e^{j\omega}) = \sum_{n=0}^{2} h[n] e^{-j\omega n}$$

$$= 1 - e^{-j2\omega} = 2 \cdot e^{-j\omega} \cdot j \cdot \sin\omega$$

$$= 2 \cdot \sin\omega \cdot e^{-j(\omega - \pi/2)}$$